Normal Distribution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Finding probabilities using the standard normal table**

A random variable $X \sim N(100, 225)$. Find $P(X < 115)$, given that $\Phi(1) \approx 0.8413$.

$\sigma = \sqrt{225} = 15$. $z = \frac{115 - 100}{15} = 1$. $P(X < 115) = \Phi(1) \approx 0.8413$.

2 **[Foundational] • Target: Properties of the normal distribution**

State two key properties of the normal distribution curve.

The normal distribution curve is symmetric (bell-shaped) about its mean μ, and the total area under the curve equals 1. (Other valid properties: mean = median = mode; the curve is asymptotic to the horizontal axis.)

3 **[Foundational] • Target: Properties of the normal distribution**

Explain what it means for the normal distribution curve to be "asymptotic" to the horizontal axis.

It means the curve approaches the horizontal axis as $x \to \pm\infty$ but never actually touches or crosses it — the probability density gets arbitrarily close to zero but is never exactly zero for any finite x.

4 **[Foundational] • Target: Properties of the normal distribution**

State one similarity and one difference between the normal distribution and the discrete uniform distribution.

Similarity: both are probability distributions in which the total probability sums (or integrates) to 1. Difference: the normal distribution is continuous and bell-shaped, with values clustering near the mean, while the discrete uniform distribution is discrete and every outcome has equal probability.

5 **[Foundational] • Target: Standardizing a normal variable**

A random variable $X \sim N(50, 64)$. Find the z-score corresponding to $x = 62$, using $z = \frac{x - \mu}{\sigma}$.

$\sigma = \sqrt{64} = 8$. $z = \frac{62 - 50}{8} = \frac{12}{8} = 1.5$.

6 **[Foundational] • Target: Standardizing a normal variable**

A random variable $X \sim N(30, 25)$. Find the z-score corresponding to $x = 23$.

$\sigma = \sqrt{25} = 5$. $z = \frac{23 - 30}{5} = \frac{-7}{5} = -1.4$.

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7 [Foundational] • Target: Standardizing a normal variable

A random variable $X \sim N(70, 144)$. Find the z-score corresponding to $x = 88$.

$\sigma = \sqrt{144} = 12$. $z = \frac{88 - 70}{12} = \frac{18}{12} = 1.5$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding probabilities using the standard normal table

Using the same distribution $X \sim N(100, 225)$, find $P(X > 130)$, given that $\Phi(2) \approx 0.9772$.

$z = \frac{130 - 100}{15} = 2$. $P(X > 130) = 1 - \Phi(2) \approx 1 - 0.9772 = 0.0228$.

9 [Intermediate] • Target: Finding probabilities using the standard normal table

A random variable $X \sim N(50, 16)$. Find $P(X < 54)$, given that $\Phi(1) \approx 0.8413$.

$\sigma = \sqrt{16} = 4$. $z = \frac{54 - 50}{4} = 1$. $P(X < 54) = \Phi(1) \approx 0.8413$.

10 [Intermediate] • Target: Finding probabilities using the standard normal table

A random variable $X \sim N(80, 100)$. Find $P(X > 95)$, given that $\Phi(1.5) \approx 0.9332$.

$\sigma = \sqrt{100} = 10$. $z = \frac{95 - 80}{10} = 1.5$. $P(X > 95) = 1 - \Phi(1.5) \approx 1 - 0.9332 = 0.0668$.

11 [Intermediate] • Target: Finding a value given a probability (inverse normal)

A random variable $X \sim N(60, 100)$. Find the value x such that $P(X < x) = 0.9332$, given that $\Phi(1.5) \approx 0.9332$.

Since $\Phi(1.5) \approx 0.9332$, the corresponding z-value is 1.5. $x = \mu + z\sigma = 60 + 1.5(10) = 75$.

12 [Intermediate] • Target: Finding a value given a probability (inverse normal)

A random variable $X \sim N(80, 64)$. Find the value x such that $P(X < x) = 0.9772$, given that $\Phi(2) \approx 0.9772$.

Since $\Phi(2) \approx 0.9772$, the corresponding z-value is 2. $x = \mu + z\sigma = 80 + 2(8) = 96$.

13 [Intermediate] • Target: Finding a value given a probability (inverse normal)

A random variable $X \sim N(45, 49)$. Find the value x such that $P(X < x) = 0.8413$, given that $\Phi(1) \approx 0.8413$.

Since $\Phi(1) \approx 0.8413$, the corresponding z-value is 1. $x = \mu + z\sigma = 45 + 1(7) = 52$.

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14 **[Intermediate]** • **Target: The probability of a value lying between two bounds**

A random variable $X \sim N(40, 25)$. Find $P(35 < X < 48)$, given that $\Phi(1) \approx 0.8413$ and $\Phi(1.6) \approx 0.9452$.

$\sigma = 5$. For $x = 35$: $z = \frac{35 - 40}{5} = -1$. For $x = 48$: $z = \frac{48 - 40}{5} = 1.6$.

$P(35 < X < 48) = \Phi(1.6) - \Phi(-1) = \Phi(1.6) - [1 - \Phi(1)] = 0.9452 - (1 - 0.8413) = 0.9452 - 0.1587 = 0.7865$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The probability of a value lying between two bounds** • **Cross-Lesson: 15.01 Representation of Data**

A random variable $X \sim N(20, 16)$. Find $P(16 < X < 26)$, given that $\Phi(1) \approx 0.8413$ and $\Phi(1.5) \approx 0.9332$.

$\sigma = 4$. For $x = 16$: $z = \frac{16 - 20}{4} = -1$. For $x = 26$: $z = \frac{26 - 20}{4} = 1.5$.

$P(16 < X < 26) = \Phi(1.5) - \Phi(-1) = \Phi(1.5) - [1 - \Phi(1)] = 0.9332 - (1 - 0.8413) = 0.9332 - 0.1587 = 0.7745$.

16 **[Challenging]** • **Target: The probability of a value lying between two bounds** • **Cross-Lesson: 16.01 Permutations & Combinations**

A random variable $X \sim N(50, 100)$. Find $P(35 < X < 65)$, given that $\Phi(1.5) \approx 0.9332$.

$\sigma = 10$. For $x = 35$: $z = \frac{35 - 50}{10} = -1.5$. For $x = 65$: $z = \frac{65 - 50}{10} = 1.5$. Since these bounds are symmetric about the mean: $P(35 < X < 65) = \Phi(1.5) - \Phi(-1.5) = 2\Phi(1.5) - 1 = 2(0.9332) - 1 = 0.8664$.

17 **[Challenging]** • **Target: The normal approximation to the binomial distribution** • **Cross-Lesson: 17.01 Discrete Random Variables**

A random variable $X \sim B(200, 0.4)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid (both np and $n(1 - p)$ should exceed 5).

$np = 200(0.4) = 80$ and $n(1 - p) = 200(0.6) = 120$, both exceed 5, so the approximation is valid. Use $X \approx N(80, 48)$, since $\text{Var}(X) = np(1 - p) = 200(0.4)(0.6) = 48$.

18 **[Challenging]** • **Target: The normal approximation to the binomial distribution** • **Cross-Lesson: 18.01 Binomial Distribution**

A random variable $X \sim B(150, 0.5)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

$np = 150(0.5) = 75$ and $n(1 - p) = 150(0.5) = 75$, both exceed 5, so the approximation is valid. Use $X \approx N(75, 37.5)$, since $\text{Var}(X) = np(1 - p) = 150(0.5)(0.5) = 37.5$.

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19 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A random variable $X \sim B(100, 0.3)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

$np = 100(0.3) = 30$ and $n(1-p) = 100(0.7) = 70$, both exceed 5, so the approximation is valid. Use $X \approx N(30, 21)$, since $\text{Var}(X) = np(1-p) = 100(0.3)(0.7) = 21$.

20 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 3.01 Graphs & Transformations

A random variable $X \sim B(250, 0.2)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

$np = 250(0.2) = 50$ and $n(1-p) = 250(0.8) = 200$, both exceed 5, so the approximation is valid. Use $X \approx N(50, 40)$, since $\text{Var}(X) = np(1-p) = 250(0.2)(0.8) = 40$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Standard Normal Distribution $Z \sim N(0, 1)$

Find $P(Z < 1.645)$ for the standard normal distribution $Z \sim N(0, 1)$.

a. 0.9500 ✓

b. 0.9750

c. 0.9000

d. 0.9900

From the standard normal cumulative distribution tables, $\Phi(1.645) = 0.9500$.

22 [Foundational] • Target: The Standard Normal Distribution $Z \sim N(0, 1)$

Find $P(Z > -1.282)$ for $Z \sim N(0, 1)$.

a. 0.9000 ✓

b. 0.1000

c. 0.8000

d. 0.8500

By symmetry, $P(Z > -1.282) = P(Z < 1.282) = \Phi(1.282) = 0.9000$.

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23 [Foundational] • Target: The Standard Normal Distribution $Z \sim N(0, 1)$

Find $P(-1.96 < Z < 1.96)$ for $Z \sim N(0, 1)$.

a. 0.9500 ✓

b. 0.9750

c. 0.9000

d. 0.9900

$P(-1.96 < Z < 1.96) = 2\Phi(1.96) - 1 = 2(0.9750) - 1 = 1.95 - 1 = 0.9500$.

24 [Foundational] • Target: Standardization & General Normal Distribution

The random variable $X \sim N(50, 16)$. Find $P(X < 56)$.

a. 0.9332 ✓

b. 0.8413

c. 0.9772

d. 0.8849

Standard deviation $\sigma = \sqrt{16} = 4$. $z = \frac{56 - 50}{4} = 1.5$. $P(X < 56) = \Phi(1.5) = 0.9332$.

25 [Foundational] • Target: Standardization & General Normal Distribution

Heights of adult females are normally distributed with mean 165 cm and standard deviation 6 cm. What percentage of females are taller than 177 cm?

a. 2.28% ✓

b. 5.00%

c. 4.56%

d. 1.14%

$z = \frac{177 - 165}{6} = 2.0$. $P(X > 177) = 1 - \Phi(2.0) = 1 - 0.9772 = 0.0228 = 2.28\%$.

26 [Foundational] • Target: Standardization & General Normal Distribution

If $Y \sim N(100, 225)$, find $P(85 < Y < 115)$.

a. 0.6826 ✓

b. 0.9544

c. 0.5000

d. 0.7500

$\sigma = \sqrt{225} = 15$. The interval $85 < Y < 115$ represents $\mu \pm 1\sigma$.

$P(-1 < Z < 1) = 2\Phi(1) - 1 = 2(0.8413) - 1 = 0.6826$.

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27 [Foundational] • Target: Inverse Normal & Finding μ and σ

The random variable $X \sim N(\mu, 25)$ has $P(X < 62) = 0.8413$. Find the mean μ.

a. **57** ✓

b. 55

c. 52

d. 60

$\Phi(z) = 0.8413 \Rightarrow z = 1.0$. Since $\sigma = 5$: $1.0 = \frac{62 - \mu}{5} \Rightarrow 62 - \mu = 5 \Rightarrow \mu = 57$.

28 [Foundational] • Target: Inverse Normal & Finding μ and σ

For a normal variable X, $P(X < 40) = 0.05$ and $P(X < 70) = 0.95$. Find μ and σ.

a. $\mu = 55, \sigma = 9.12$ ✓

b. $\mu = 55, \sigma = 15.0$

c. $\mu = 50, \sigma = 10.0$

d. $\mu = 55, \sigma = 5.50$

By symmetry, $\mu = (40 + 70)/2 = 55$. For 95th percentile, $z = 1.645$: $1.645 = \frac{70 - 55}{\sigma} \Rightarrow \sigma = \frac{15}{1.645} \approx 9.12$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Inverse Normal & Finding μ and σ

Packets of biscuits have nominal weight 200 g. Actual weights are normally distributed with $\mu = 204$ g and $\sigma = 2.5$ g. What proportion of packets are underweight (< 200 g)?

a. 0.0548 ✓

b. 0.0228

c. 0.0100

d. 0.0808

$z = \frac{200 - 204}{2.5} = -1.6$. $P(X < 200) = \Phi(-1.6) = 1 - \Phi(1.6) = 1 - 0.9452 = 0.0548$.

30 [Intermediate] • Target: Normal Approximation to the Binomial Distribution

State the two conditions required to approximate a binomial distribution $X \sim B(n, p)$ by a normal distribution.

a. $np > 5$ **and** $n(1 - p) > 5$ ✓

b. $n > 30$ and $p = 0.5$

c. $np > 10$ only

d. $n(1 - p) > 100$

The standard Cambridge AS Level guideline requires both $np > 5$ and $n(1 - p) > 5$ for an adequate normal approximation.

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31 [Intermediate] • Target: Normal Approximation to the Binomial Distribution

A fair coin is tossed 100 times. Using the normal approximation with continuity correction, what is the expression for $P(X \geq 55)$?

a. $P(Y > 54.5)$ **where** $Y \sim N(50, 25)$ ✓

b. $P(Y > 55.5)$ where $Y \sim N(50, 25)$

c. $P(Y > 54.5)$ where $Y \sim N(50, 5)$

d. $P(Y > 55.0)$ where $Y \sim N(50, 25)$

$\mu = 100(0.5) = 50$, $\sigma^2 = 100(0.5)(0.5) = 25$. For $X \geq 55$, the continuity correction shifts to the left boundary: $P(Y > 54.5)$.

32 [Intermediate] • Target: Normal Approximation to the Binomial Distribution

Compute the approximate probability of getting at least 55 heads in 100 tosses using $Y \sim N(50, 25)$.

a. 0.1841 ✓

b. 0.1587

c. 0.2119

d. 0.1250

$z = \frac{54.5 - 50}{5} = 0.90$. $P(Y > 54.5) = 1 - \Phi(0.90) = 1 - 0.8159 = 0.1841$.

33 [Intermediate] • Target: Properties of the normal distribution

Given $X \sim N(\mu, \sigma^2)$, $P(X < 57) = 0.9$ and $P(X < 27) = 0.1$, find μ, correct to the nearest integer.

a. 39

b. **42** ✓

c. 48

d. 45

Using $z_1 = 1.2816$, $z_2 = -1.2816$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 42$.

34 [Intermediate] • Target: Standardizing a normal variable

Given $X \sim N(\mu, \sigma^2)$, $P(X < 100) = 0.95$ and $P(X < 80) = 0.05$, find μ, correct to the nearest integer.

a. 87

b. 93

c. 96

d. **90** ✓

Using $z_1 = 1.6449$, $z_2 = -1.6449$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 90$.

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35 [Intermediate] • Target: Finding probabilities using the standard normal table

Given $X \sim N(\mu, \sigma^2)$, $P(X < 51) = 0.8$ and $P(X < 37) = 0.2$, find μ, correct to the nearest integer.

a. 50

b. 41

c. 47

d. **44** ✓

Using $z_1 = 0.8416$, $z_2 = -0.8416$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 44$.

36 [Intermediate] • Target: Finding a value given a probability (inverse normal)

Given $X \sim N(\mu, \sigma^2)$, $P(X < 46) = 0.8$ and $P(X < 32) = 0.2$, find μ, correct to the nearest integer.

a. 45

b. **39** ✓

c. 42

d. 36

Using $z_1 = 0.8416$, $z_2 = -0.8416$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 39$.

37 [Intermediate] • Target: The probability of a value lying between two bounds

Given $X \sim N(\mu, \sigma^2)$, $P(X < 107) = 0.9$ and $P(X < 77) = 0.1$, find μ, correct to the nearest integer.

a. **92** ✓

b. 98

c. 89

d. 95

Using $z_1 = 1.2816$, $z_2 = -1.2816$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 92$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 18.01 Binomial Distribution

Given $X \sim N(\mu, \sigma^2)$, $P(X < 55) = 0.9$ and $P(X < 29) = 0.1$, find μ, correct to the nearest integer.

a. 45

b. **42** ✓

c. 39

d. 48

Using $z_1 = 1.2816$, $z_2 = -1.2816$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 42$.

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39 [Challenging] • Target: Solving for unknown parameters of a normal distribution • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Given $X \sim N(\mu, \sigma^2)$, $P(X < 36) = 0.85$ and $P(X < 12) = 0.15$, find μ, correct to the nearest integer.

a. 27

b. **24** ✓

c. 30

d. 21

Using $z_1 = 1.0364$, $z_2 = -1.0364$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 24$.

40 [Challenging] • Target: A well-designed word problem: setting a pass mark for an exam • Cross-Lesson: 3.01 Graphs & Transformations

Given $X \sim N(\mu, \sigma^2)$, $P(X < 103) = 0.9$ and $P(X < 77) = 0.1$, find μ, correct to the nearest integer.

a. 96

b. 87

c. 93

d. **90** ✓

Using $z_1 = 1.2816$, $z_2 = -1.2816$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 90$.

41 [Challenging] • Target: The Standard Normal Distribution $Z \sim N(0, 1)$ • Cross-Lesson: 4.01 Coordinate Geometry

Given $X \sim N(\mu, \sigma^2)$, $P(X < 99) = 0.8$ and $P(X < 83) = 0.2$, find μ, correct to the nearest integer.

a. 94

b. **91** ✓

c. 88

d. 97

Using $z_1 = 0.8416$, $z_2 = -0.8416$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 91$.

42 [Challenging] • Target: Standardization & General Normal Distribution • Cross-Lesson: 5.01 Trigonometry

Given $X \sim N(\mu, \sigma^2)$, $P(X < 66) = 0.9$ and $P(X < 36) = 0.1$, find μ, correct to the nearest integer.

a. 48

b. 57

c. 54

d. **51** ✓

Using $z_1 = 1.2816$, $z_2 = -1.2816$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 51$.

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43 [Challenging] • Target: Inverse Normal & Finding μ and σ • Cross-Lesson: 6.01 Differentiation & Integration

A random variable $X \sim B(201, 0.75)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.

a. 148.8

b. 152.8

c. 155.8

d. **150.75** ✓

Applying Normal Distribution principles, Mean $= np = 201 \times 0.75 = 150.75$ Therefore, the verified correct option is 150.75.

44 [Challenging] • Target: Normal Approximation to the Binomial Distribution • Cross-Lesson: 7.01 Statistics & Probability

A random variable $X \sim B(266, 0.7)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.

a. 184.2

b. **186.2** ✓

c. 188.2

d. 191.2

Applying Normal Distribution principles, Mean $= np = 266 \times 0.7 = 186.2$ Therefore, the verified correct option is 186.2.

45 [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 8.01 Functions: Composite & Inverse

A random variable $X \sim B(170, 0.25)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.

a. 44.5

b. 47.5

c. **42.5** ✓

d. 40.5

Applying Normal Distribution principles, Mean $= np = 170 \times 0.25 = 42.5$ Therefore, the verified correct option is 42.5.

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